

Short automated extensions

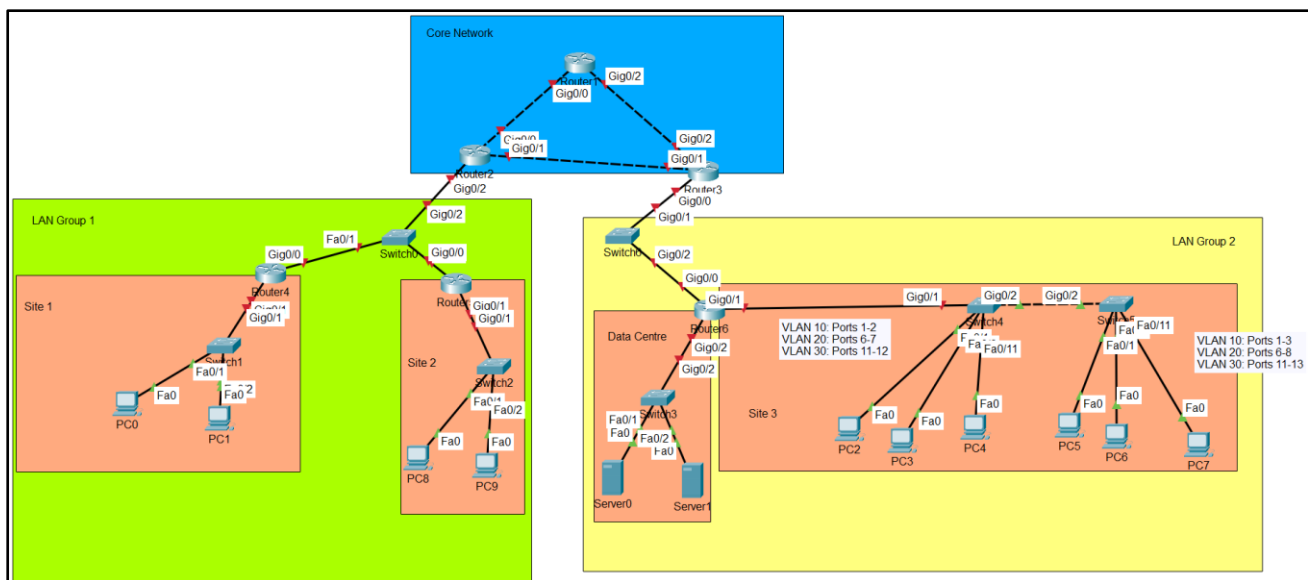
Yes. Automatic short extension is available, apply before the due date via Service Connect Portal

Assessment Overview

Students will configure a multi-site enterprise network in Cisco Packet Tracer for an organisation operating across multiple geographic locations with a centralized Data Centre as a network engineer would for a real client. Please note that the network topology we are working on for this project represents only a portion of the organisation's overall network, not the complete infrastructure. Please also note that some parts of the topology could have been designed more efficiently (e.g., Site 3), however, to keep things simple and closely aligned with your practical/workshop work, they have been intentionally left as they are. There are two deliverables for this assessment: a configured Packet Tracer file and a design report.

Assessment Details / Specific Tasks

Personalized specifications have been provided to every student. Please check the *Assessment 2 – Project* section on your iLearn unit page. A packet tracer file has been provided containing all required devices and cabling, however, no configurations have been applied. The physical layout has been organized, colour coded, and labelled: Core Network, LAN Group 1, LAN Group 2, Site 1, Site 2, Site 3, Data Centre (Refer to the screenshot below). Please note that the device name labels and port labels have been made visible by choosing preferences (interface) from the options menu at the top left. You may choose to make them invisible if needed. Also, please note that only a limited number of hosts (PCs/Servers) have been connected to each network for connectivity testing purposes.



Task 1: Perform VLSM Subnetting (05 marks)

The address block available to you has been provided in the personalised specifications under the *Assessment 2 – Project* section. You are required to perform VLSM subnetting on the provided address block according to the number of hosts per site/network or link given below. Please note that all the router interfaces have already been included in the following hosts. Also, none of the switches will need an IP address so we will not be considering them. You

must keep the subnet size to minimum unless advised otherwise. The entire VLSM process must be added to the design report (refer to the deliverables' section).

- i. Site 1: Refer to the personalised specifications under the *Assessment 2 – Project* section
- ii. Site 2: 24 hosts
- iii. Site 3: 3 VLANs with 5 hosts each
- iv. Data Centre: 50 hosts
- v. Point to point link Router1-Router2
- vi. Point to point link Router1-Router3
- vii. Point to point link Router2-Router3
- viii. Router2-Router4-Router5 connecting network
- ix. Router3-Router6 connecting network (Please think about the subnet length carefully to leave room for future scalability (up to 5 hosts in total))

Task 2: Assign Hostnames to all Routers and Switches (no marks)

The hostnames should follow a consistent naming convention, e.g., Core-Router1, 2, 3, Switch-Site1, 2,..., Switch-LAN1, 2,..., Switch-DC. Hostnames do not need to be assigned to end-user PCs or servers, they will be referred to by their default names. This task carries no individual marks, however, completing it is strongly recommended as a best practice activity.

Task 3: Configure Router Interfaces (03 marks)

Each active interface of the router must be configured with the correct IP address and subnet mask based on the VLSM calculations for the connected network. Please note that assigning IPs to switches is not a part of this project. There is no need to assign IP to the router interface connected to Site 3.

Task 4: Configure PCs and Servers (02 marks)

Assign IPs, subnet masks and default gateways to all PCs and servers based on the network they are a part of and the connected router configurations. For PCs on Site 3, there is no need to assign the default gateway as we will not be implementing interVLAN routing.

Task 5: Configure Static Routes (03 marks)

Enable inter-site communication by configuring appropriate static routes between the networks. Site 1 should be able to communicate with Site 2 and the data centre. Similarly, Site 2 should be able to communicate with Site 1 and the data centre. Please note that you will not be able to communicate with PCs on Site 3.

Task 6: Configure VLANs (03 marks)

Site 3 must be configured with 3 VLANs: VLAN 10, VLAN 20, and VLAN 30. Each VLAN has 5 hosts as mentioned above and 5 ports (in total) must be assigned to each VLAN, i.e., VLAN 10 must have 5 ports in total, i.e., ports 1-2 on the switch (i.e., switch on the left hand side) must be allocated to VLAN 10 and ports 1-3 on the other switch (i.e., switch on the right hand side) must also be allocated to VLAN 10, similarly, VLAN 20 must have 5 ports in total, i.e., ports 6-7 on the switch (i.e., switch on the left hand side) must be allocated to VLAN 20 and ports 6-8 on the other switch (i.e., switch on the right hand side) must be allocated to VLAN 20, and VLAN 30 must also have 5 ports in total, i.e., ports 11-12 on the switch (i.e., switch on the left hand side) must be allocated to VLAN 30 and ports 11-13 on the other switch (i.e., switch on the right hand side) must also be allocated to VLAN 30. As mentioned above, we will not be

implementing interVLAN routing. (*Hint: To transport data from multiple VLANs over a single link, make sure the correct modes have been enabled on the switchports*)

Task 7: Test and Verify Intra and Inter Network Connectivity (05 marks)

Test connectivity by sending ping packets between hosts according to the table given below. Please note that the Device label names used in the table below are according to the provided packet tracer file and not the hostnames. Please complete the table with IP addresses and ping pass/fail results and include in the design report (refer to the deliverables' section).

Source	Source IP Address	Destination	Destination IP Address	Pass/Fail
PC0		PC1		
PC0		PC8		
PC0		Router 1 Gig0/0		
PC8		Router 1 Gig0/2		
PC0		Server1		
PC9		Server0		
PC2		PC5		
PC3		PC6		
PC4		PC7		
PC2		PC3		

Task 8: Answer the following questions (04 marks)

Each answer should be no more than 50 words. The answers must be included in the design report (refer to the deliverables' section).

- The organization plans to expand by adding two new sites, what physical topology options, changes, or additions could be considered for the network design? Please answer with reference to the provided network topology and include a diagram for your suggested network topology.
- Explain why VLSM is a better choice for this network instead of fixed size subnets?
- What is the role of the default gateway (configured on a PC) during communication?
- What is the relationship between virtualization and AWS in cloud computing?

Deliverables

Your submission must include:

- Packet Tracer file (StudentID_PT.pkt): The final configured .pkt file. Configure the provided Starter Packet Tracer file according to Tasks 1-6 and test connectivity according to Task 7.
- Design Report (StudentID_Report.pdf): The design report must include the following:
 - Title page: Unit Code, Student Name, Student ID.
 - Client requirements summary
 - VLSM subnetting process
 - Configuration summary including key CLI commands used for configuration and verification (only include for any one router and any one switch from site 3)
 - Testing and verification table from Task 7 along with properly labelled screenshots of the results.
 - Answers to the questions in Task 8.
 - Appendix: Full addressing table for all router interfaces, PCs, and servers.

- ix. Formatting: include clear headings/structure and ensure the screenshots are readable.
- x. Maximum page limit: 15 pages

Important Note

You must use the exact values (address block and no. of hosts for Site 1) assigned to you in the personalised specifications under the *Assessment 2 – Project* section on the iLearn unit page. Each student has a different set of specifications, and using incorrect data or another student's specifications may result in a failing grade.

Quality Criteria

A high-quality submission is:

- Well-structured with clear sections and logical flow
- Accurate and detailed addressing information
- Verifiable results, supported by clear tables and screenshots
- Clear, concise, and professional writing, error free
- Proper referencing of highly relevant literature (if any)
- Directly meets task requirements and demonstrates learning outcomes.

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The rubric below guides you to what we expect.

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Task 1 - VLSM (05 marks)	Correctly uses assigned host counts and network block. VLSM design clearly justified based on the provided data. All calculations are correct.	Correctly uses assigned host counts and network block. minor issues with VLSM justifications based on the provided data. 1-2 calculation errors.	Correctly uses assigned host counts and network block. minor issues with VLSM justifications based on the provided data. 3-4 calculation errors.	Correctly uses assigned host counts and network block. minor issues with VLSM justifications based on the provided data. 5-6 calculation errors.	Incorrect network block. Incorrect host counts. major issues with VLSM justifications based on the provided data. More than 6 calculation errors. Used another student's data.
Task 3 – Router Configuration (03 Marks)	All router interfaces correctly configured.	Almost all router interfaces correctly configured. 1-2 minor errors.	Most router interfaces correctly configured. 3-4 minor errors.	Many router interfaces correctly configured. 5-6 minor errors.	Router interfaces incorrectly configured. More than 6 errors.
Task 4 – PCs' and Servers' Configuration (02 Marks)	All PCs and Servers correctly configured.	Almost all PCs and Servers correctly configured. 1-2 minor errors.	Most PCs and Servers correctly configured. 3-4 minor errors.	Many PCs and Servers correctly configured. 5-6 minor errors.	PCs and Servers incorrectly configured. More than 6 errors.
Task 5 – Routing (03 Marks)	All routes correctly configured.	Almost all routes correctly configured. 1-2 minor errors.	Most routes correctly configured. 3-4 minor errors.	Many routes correctly configured. 5-6 minor errors.	Routes incorrectly configured. More than 6 errors.
Task 6 – VLANs (03 Marks)	All VLANs correctly configured. Correct port modes. Correct port allocations.	All VLANs correctly configured. Minor error in port modes. Minor error in port allocations.	2 VLANs correctly configured. Minor error in port modes. Minor error in port allocations.	1 VLAN correctly configured. Minor error in port modes. Minor error in port allocations.	VLANs incorrectly configured. Incorrect port modes. Incorrect port allocations.
Task 7 – Testing (05 Marks)	All required tests completed with correct pass/fail status. Table completed with IPs and results. All screenshots included.	Almost all required tests completed with correct pass/fail status. Table completed with IPs and results. All screenshots included. 1-2 minor issues.	Most of the required tests completed with correct pass/fail status. Table completed with IPs and results. All screenshots included. 3-4 minor issues.	Many of the required tests completed with correct pass/fail status. Table completed with IPs and results. All screenshots included. 5-6 minor issues.	Incomplete testing. Incomplete table. Missing screenshots. More than 6 minor issues.

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Task 8 – Short Answers (04 Marks)	All 4 questions answered perfectly. Word count within 50 words.	3 questions answered perfectly. Word count within 50 words.	2 questions answered perfectly. Word count within 50 words.	1 question answered perfectly. Word count within 50 words.	Missing. Word count limit exceeded for all 4 answers.
Appendix – Addressing Table (02 Marks)	Complete table included with all routers, PCs, and Servers and corresponding IP configurations.	Table included with almost all routers, PCs, and Servers and corresponding IP configurations. 1-2 minor issues.	Table included with most of the routers, PCs, and Servers and corresponding IP configurations. 3-4 minor issues.	Table included with many of the routers, PCs, and Servers and corresponding IP configurations. 5-6 minor issues.	Table not included. More than 6 minor issues.
Task 4 – Report Quality (03 Marks)	Complete, concise, well-structured and polished. All sections present. All tables are clear. All screenshots are properly labelled and readable.	Complete, well-structured and polished with 1-2 minor issues with tables/screenshots labelling/ clarity.	Complete, well-structured and polished with 3-4 minor issues with tables/screenshots labelling/ clarity.	Complete, well-structured and polished with 5-6 minor issues with tables/screenshots labelling/ clarity.	Incomplete, poorly organized, more than 6 minor issues with tables/screenshots labelling/ clarity.